

Kavindu Munugoda (PhD)

Pharmacognosy | Phytochemistry | Molecular Virology | Ethnobotany

✉ kmunugoda@gmail.com ☎ [+94 77 780 6927](tel:+94777806927) 📍 Ganemulla, Sri Lanka 🌐 [Portfolio](#) [LinkedIn](#) [ORCID](#)

Professional Profile

Ph.D. qualified Research Associate and Natural Products Scientist currently engaged at the Medical Research Institute (MRI), Sri Lanka. Offers over 6 years of cumulative experience in tertiary education and high-impact biomedical research. Proven expertise in bridging traditional ethnobotanical knowledge with modern viral therapeutic development, specifically targeting HPV and vector-borne diseases. Validated history of **delivering undergraduate curriculum**, **supervising Honours research students**, and **securing competitive international funding** (ICGEB).

Key Competencies

💡 Drug Discovery & Computational Biology

genome assembly and annotation, Molecular docking and in silico screening, protein-ligand interaction analysis, structure-based drug design, PyRx, PyMOL, UCSF ChimeraX, AlphaFold

🔬 Analytical Chemistry & Natural Products

GC-MS analysis, UV spectroscopy, hydrodistillation, Soxhlet extraction, phytochemical screening and characterization, Micro-Kjeldahl analysis, bioactive compound isolation

🧬 Molecular Biology & Protein Biochemistry

Recombinant DNA technology, molecular cloning and plasmid construction, PCR and gel electrophoresis, CTAB genomic DNA extraction, recombinant protein expression (E. coli), GST-affinity chromatography, SDS-PAGE, Western blotting

🧫 Cell-Based Assays & Virology

Mammalian cell culture across multiple lines (HEK293T, HaCaT, HeLa, CaSki, C33A, SiHa), cytotoxicity profiling and IC₅₀ determination (CCK-8 assay), dual-luciferase reporter systems, cell transfection, viral protein expression and functional characterization

🦠 Microbiology & Bioprocessing

Microbial isolation and identification, colony characterization, aseptic technique and subculture, media preparation, fermentation processes, quality control microbiology

📊 Bioinformatics & Data Science

Statistical analysis and modeling (Python, R, GraphPad Prism, IBM SPSS, Minitab), dose-response curve fitting (4PL models), multivariate analysis (PCA, logistic regression, Random Forest), genomic data analysis (NOVOPlasty, MAFFT)

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Recent Academic & Research Employment

Department of Vaccine QC and Molecular Biology
Research Associate

Medical Research Institute (MRI) – Ministry of Health, Sri Lanka
Jan 2026 – Present

- Instruction & Training:** Conduct comprehensive training sessions for junior staff on advanced molecular biology techniques, including dual-luciferase reporter assays and cell culture methodologies.
- Assessment Design:** Evaluate technical competency assessments and laboratory practical examinations for training programs to ensure staff meet research standards.
- Student Mentorship:** Mentor visiting students and junior researchers on experimental design, troubleshooting, and career development in molecular virology.
- Research Leadership:** Supervise comprehensive research workflows for junior staff, coordinating experimental planning and milestone tracking.

International Centre for Genetic Engineering and Biotechnology China
Regional Research Centre (ICGEB China RRC)

Medical High-tech Zone, Taizhou, China

Visiting Scientist

December 2024 - December 2025

- Instruction & Training:** Facilitated technical training sessions for laboratory personnel on advanced molecular biology techniques, including molecular docking, cell culture methodologies, and protein expression systems.
- Research Innovation:** Designed and executed comprehensive research investigating novel plant-based antiviral therapeutics targeting HPV viral replication machinery.
- Strategic Collaboration:** Served as Delegation Officer for bilateral research visits and facilitated the establishment of formal research partnerships between China Medical City and the University of Kelaniya.

Education

Faculty of Graduate Studies
Pharmacognosy & Phytochemistry

Doctor of Philosophy (PhD)
University of Sri Jayewardenepura, Sri Lanka • 2023

Thesis: *Production of Mosquito Repellents Using Lesser Known, Under-Utilized Volatile Oil Producing Plant Species.*

Faculty of Applied Sciences
Plant Biotechnology

Bachelor of Science (Hons) • Second Class
University of Sri Jayewardenepura, Sri Lanka • 2018

Thesis: *Evaluation of Microbial Counts in Apple Flavoured Beer Compared to Base Beer after Exposing to an External Environment.*

Research Funding & Awards

ICGEB-DIC-MOST International Fellowship
Postdoctoral Grant in ICGEB China RRC

2024 NSF SUSRED Award
Support Scheme for Supervision of Research Degrees

2024

Best Oral Presentation
26th International Forestry and Environment Symposium

2022 NRC Grant 18-032
Research Assistant funding stream

2019

Publications

Patents

Munugoda, K. D., Subasinghe, S., Hapugoda, M. (2024). *Volatile Plant Oil-Based Mosquito Repellent Sticks & Sprays*. National Intellectual Property Office, Sri Lanka (App No: LK/P/1/23236).

Selected Journal Articles

- Munugoda, K. D., et al. (2023). Choice modelling for the commercial cultivation of underutilized aromatic plants for producing mosquito repellents. *Economic Botany*, 1-19.
- Chen, J., ..., Munugoda, K. D., et al. (2021). The complete chloroplast genome of Walla Patta, *Gyrinops walla* (Thymelaeaceae). *Mitochondrial DNA Part B*, 6(6), 1699-1701.

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Certifications

Science, Technology and Innovation Policy and Policy Instruments for Sustainable Development Goals for Asia and the Pacific
UN Interagency Task Team on STI for the SDGs (IATT)

November 2024

PacMOSSI-Aedes & Anopheles Surveillance & Control
James Cook University, Australia

June 2024

Thermo Fisher Scientific - Mass Spectrometry Job Simulation
Forage

January 2024

Research Methodology and Scientific Writing
Faculty of Graduate Studies, University of Sri Jayewardenepura, Sri Lanka

November 2020

Memberships

Associate Member (A.I.Chem.C.)
Institute of Chemistry Ceylon

Membership No: 3548 Aff

English Language Proficiency

IELTS

Overall Band Score 7.5

References

Prof. BOON Siaw Shi
Department of Microbiology,
The Chinese University of Hong Kong,
Hong Kong SAR, China
Phone: +85 25 192 4602
Email: boonss@cuhk.edu.hk

Prof. Menaka Hapugoda
Molecular Medicine Unit,
Faculty of Medicine,
University of Kelaniya, Sri Lanka
Phone: +94 71 445 8606
E-mail: menakaha@kln.ac.lk

Dr. Dhanushka Hettiarachchi
School of Allied Health, University of
Western Australia
Phone: +61 41 979 3136
E-mail: ghanushka.hettiarachchi@uwa.edu.au